

Maximizing the Magic and Minimizing the Wait at Walt Disney World

CSC-475 Final Report

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Abstract

A trip to Disney World is a dream vacation for many worldwide. With four theme parks and two water parks filled with attractions, shows, restaurants, and shops, planning this magical experience can be challenging, especially considering the high costs and long wait times associated with "the most magical place on earth". Given the factors that influence vacation planning, such as wait times and reservations, the creation of a data visualization tool that would allow users to compare monthly hourly average wait times for individual rides, would aid greatly in planning. This project proposes the creation of a data visualization tool that enables users to compare wait times based on individual rides. Location and wait time data was collected for 16 rides from February 2023- January 2024. After analyzing the data, the average hourly wait time per month was calculated for each ride, the dashboard was designed, and then created. The Disney Wait Time Dashboard includes a filterable map of the rides, overall and monthly wait time distributions, the busiest and slowest times to go, and a tool where users can compare rides wait times.

Introduction

The goal of this project is to create a website where users can compare wait times for popular rides at Walt Disney World in Orlando Florida. With average monthly wait times ranging from 30 to over 45 minutes and individual ride wait times varying from 5 to over 100 minutes, strategic timing is crucial ¹. Selecting not only the right time to visit Disney World but also the time of day for specific rides is key to enhancing your overall experience.

Currently, Walt Disney World has the My Disney Experience app, which displays real-time wait times for rides at all theme parks, but lacks data on average wait times per ride. Given the various factors that influence vacation planning, such as wait times and reservations, the creation of a data visualization tool that would allow users to compare monthly hourly average wait times for individual rides, would be great for planning purposes.

This project aims to answer the following questions:

¹ThrillData site: <https://www.thrill-data.com/waits/chain/wdw>

1. Are there hourly trends for wait times on different rides?
2. Can using wait time data to determine when to go on a ride optimize one's trip to Disney World?

Having a tool like this could potentially ease the planning and maximize one's trip to Disney World. This project hopes to find monthly wait times trends on specific rides based on time of day.

Related Work

Theme parks are a way for many to escape reality. With rides of all types, parades, shows, sightseeing, and shopping, there are so many different things to do at theme parks that it might seem impossible to fit in everything that one may want to do. However, when going to a theme park, one unavoidable experience is waiting in line – and waiting long. With popularity of theme parks continuing to increase, waiting in line is only becoming a bigger part of the theme park experience. Over 17 million people visited Walt Disney World's Magic Kingdom in Orlando, Florida in 2022 [1], and with popularity also comes crowds and heavy traffic.

While people who visit theme parks tend to think that the parks are more crowded than not, guests associate the parks busyness with more positive words, such as “popular”, “attracts many visitors”, and “would come again”, rather than negative words such as “cramped” or “too busy”. [2] These positive attitudes are not because the parks are not busy, in fact, these theme parks are very busy. Back in early 2023, the Star Wars: Rise of the Resistance ride in Walt Disney World's Hollywood Studios had a reported wait time of 320 minutes– over five and a half hours to ride a ride [3]. This ride example illustrates the scale of demand that theme parks have, and the commitment that theme park guests have to experience a ride.

So why would someone choose to wait five and a half hours to ride a ride? Guests spend roughly the same amount of time waiting in line as they do in experiencing all activities that theme parks have to offer. [4] While park visitors attempt to be mindful of waiting times, visitors prioritize attractions with higher experience value, and these high experience value rides have longer wait times. [5] The Star Wars: Rise of the Resistance is a prime example of this; the thrill of an attraction dominates the inconvenience of waiting. In addition to the thrill of the ride, themed attractions in general have higher ratings and experience value than rides with no theme, and this is because of “the sense of immersion, memorable and the unique experiences” that the rider gets when on the ride. [6]

For many theme parks, the primary focus is the overall guest's satisfaction; that is, was their trip worth the money and did their trip meet their expectations. [7] The visitor's experience at a theme park plays a large factor on whether or not they will go back, and overall guests are satisfied. In fact, over 90 percent of returning guests and over 85 percent of new guests left theme parks satisfied and believed their trip was worth the costs. [8] Additionally, those who spend more time in the theme parks tend to feel more satisfied with their trip compared to those who spent less time at a park, and this can be from the feeling of getting the most "bang for your buck". [8] Longer trips are even shown to leave a positive impact on a guest's overall visit, as theme park visitors who stay overnight at the theme park hotels and believe their trip was a good value are reported to have greater emotional stimulation and be more satisfied with their trip. [9]

There are many different factors that contribute to the overall satisfaction of theme park visitors, such as the environment and emotions, but even the crowds that come along with high experience value attractions positively impact the overall customer satisfaction, and behaviors in a theme park. [2] While crowds might seem like a negative, visitors would rather wait in a long line and ride a popular, "better" ride than go on multiple lower experience valued rides with lower waits. Additionally, being surrounded by others who are having fun positively impacts the visitor's behavior, which again, contributes to the overall customer satisfaction of a theme park. [10] For example, one can typically expect to have more fun on a ride when everyone else is having fun and being expressive of their enjoyment, rather than no expressions at all. This is not to say that visitors enjoy waiting in lines- this is far from the truth. Long wait times rank as the second most significant factor of customer dissatisfaction in theme parks (pricing being the first), but the experience of the ride is more memorable than the wait which in turn leaves the guest overall satisfied instead of overall unsatisfied [11].

This project does not assess the efficiency in which attractions are operating, rather this project is to help create a tool to help get the most out of a costly - both time and money costly- trip to a theme park. While some level of efficiency is crucial to keep theme parks running and guests satisfied, a study done at Samsung's Everland park concluded that the rides with lower efficiency tend to have higher guest satisfaction. [12] This can be because of numerous different factors, but a probable major reason is that thrill and roller coasters tend to have lower efficiency, and as mentioned above, these attraction types have high experience value. [12] Consider our Star Wars: Rise of the Resistance ride as another example. In 2022, this top Walt Disney World attraction closed 580 times! [13]

This shows the demand for theme parks, and that if there is a specific ride that a person wants to go on, they are most likely going to wait. However, this can be a problem if a person wants to go on multiple high experience attractions – all with long lines. The purpose of this project is to create a tool that identifies the best times of year and even times of day to go on various rides in Walt Disney World. While there are existing platforms that display past wait times and average wait times for different rides, these platforms are not the most user friendly. Additionally, there is currently no platform where the user can select a month and view the corresponding hourly average wait times for individual rides. This tool will help visitors plan their trip, which in turn will optimize one's trip.

Park	Ride
Animal Kingdom	Avatar Flight Passage
Animal Kingdom	Avatar Na'Vi River
Animal Kingdom	Expedition Everest
Animal Kingdom	Kilimanjaro Safaris
Epcot	Frozen Ever After
Epcot	Remy's Ratatouille Adventure
Epcot	Test Track
Hollywood Studios	Slinky Dog Dash
Hollywood Studios	Rise Of Resistance
Hollywood Studios	Mickey Minnie Runaway Railway
Hollywood Studios	Tower Of Terror
Magic Kingdom	Space Mountain
Magic Kingdom	Seven Dwarfs Mine Train
Magic Kingdom	Big Thunder Mountain
Magic Kingdom	Jungle Cruise
Magic Kingdom	Peter Pan's Flight

Table 1: 16 rides from Walt Disney World's four theme parks used for this project.

Data

Data was collected from ThrillData ², a site that tracks wait times from numerous theme parks worldwide. Prior to February 2024, Thrill Data had historic wait times from the past year for most rides in Walt Disney World publicly available, but has since removed that data from their site. The removal of this data occurred during the collection of this project, which limited the number of rides available on the dashboard.

This project was able to collect wait time information from February 2023 to January 2024 for 16 rides in Walt Disney World. The rides that were chosen to be included in this project were among the top five busiest rides in the park that they are located in and have traditional queues with average wait times above 45 minutes, which is roughly the overall average wait time in Walt Disney World. This means that rides with virtual line queues, such as Epcot's Guardians of the Galaxy and Magic Kingdom's TRON Lightcycle Run, were not included. Hollywood Studios' Rock'n'Rollercoaster was not included because it was closed, and still is closed, for refurbishment. Additionally, this project wanted to include Hollywood Studios' Toy Story Mania and Epcot's Soarin, but was unable to because of the data removal from ThrillData.

The data for each ride from ThrillData included the ride name, date, time, and the wait time at the given time. There were wait times for roughly every minute of the day that the ride was in operation. That is, if it was during park hours but the ride was closed, there was no wait time recorded. For each month, average wait times every open hour for each ride were computed, and then put into an overall sheet for analysis. Below is an example of the data for Magic Kingdom's Seven Dwarfs Mine Train ride in February 2023, and this was done for every month for each ride. Additionally, daily wait time averages were computed to find busiest holidays, and location data was manually collected for each ride. Out of the 16 rides, monthly data was available for each ride except for Magic Kingdom's Jungle Cruise in December 2023 since the ride was closed during that month.

²ThrillData site: <https://www.thrill-data.com>

Seven Dwarfs Mine Train: February 2023	
Time Of Day	Average Wait Time (in Minutes)
12 AM	28
7 AM	64
8 AM	67
9 AM	70
10 AM	78
11 AM	94
12 PM	85
1 PM	87
2 PM	92
3 PM	89
4 PM	89
5 PM	86
6 PM	84
7 PM	75
8 PM	60
9 PM	72
10 PM	48
11 PM	36
Grand Total	77

Table 2: Walt Disney World's Seven Dwarfs Mine Train Ride in Magic Kingdom. Average wait time in February 2023.

Methods

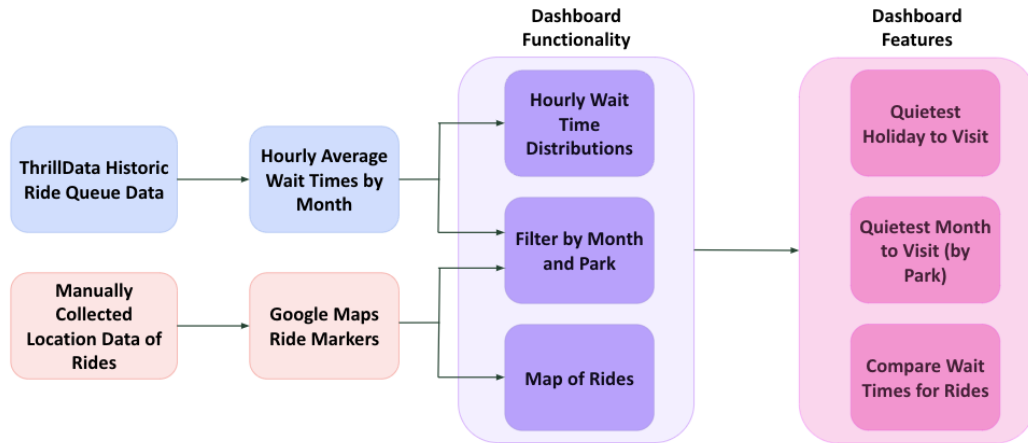


Figure 1: Pipeline of the Project. First, data is collected to get necessary data for dashboard functionality. Then, the basic functionality of the dashboard will also include some features.

Figure 1 shows the process of this project. As mentioned in the **Data** section, location data and historic wait time data was collected for the 16 rides shown in Table 2. Location data was collected for each attraction to create a marker on a map for it. This allows the user to see which rides are close together, as well as the ability to filter attractions based on parks. Historic wait time data was used to compute average hourly wait times of each ride overall and by month. This allowed for hourly wait time distributions for each ride and the ability to filter the wait times by month allowed for users to get a more accurate representation of what a wait for a specific ride would be at a month of visit.

Some features of the dashboard include quietest/busiest months to visit, quietest/busiest holidays at Disney World, and the ability to compare wait times. To find the quietest/busiest months to visit Disney World, the overall average wait time for each park was calculated to determine the slowest and busiest months at each park, and then the average wait time for all attractions was calculated to find the overall slowest and busiest months at Walt Disney World. For the busiest and quietest holidays, national holidays were identified and daily average wait times for all rides were calculated for the each holiday. Additionally, the feature of comparing wait times of rides allows users to choose a specific month, and select rides.

Finding the hourly averages for each month was used using Microsoft Excel ³, analysis was done using the Pandas library in Python⁴, and lastly the implementation of the dashboard and creation of the dashboard was used using Javascript with HTML/CSS ⁵. To view analysis, go to the project's github page: <https://github.com/abbydumke/CS475Project>.

Results

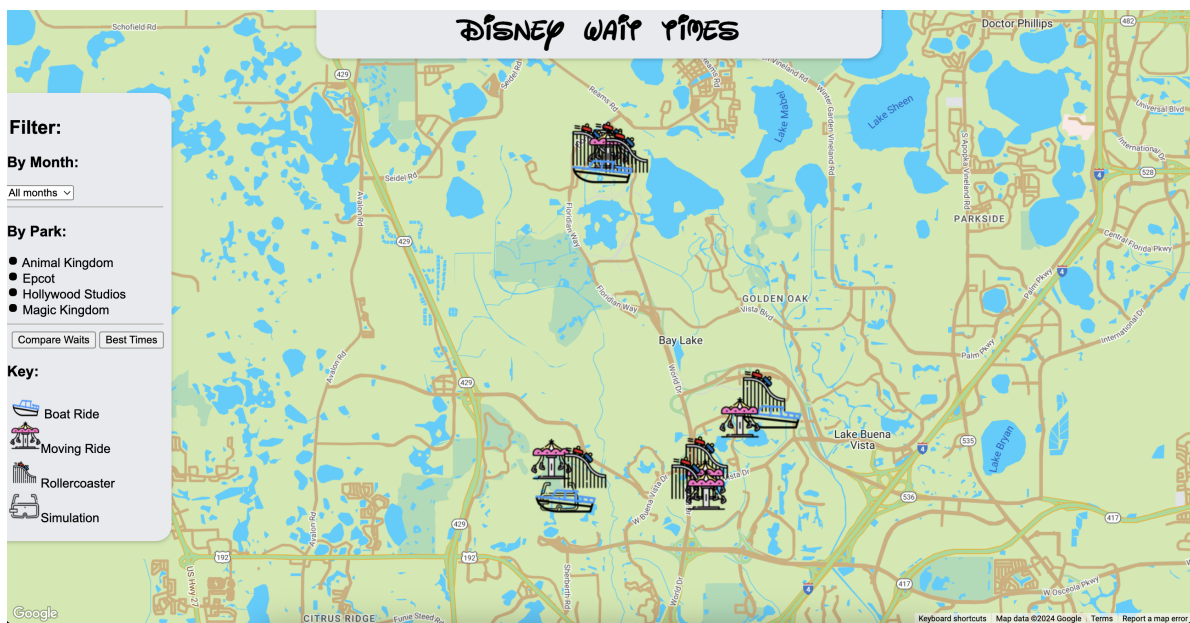


Figure 2: Home screen of the dashboard.

Figure 2 shows the home page of the dashboard, which can be accessed at <https://abbydumke.github.io/CS475Project/>. The left side of the page has a sidebar which allows to filter rides based on month, parks, a key to the ride markers based on ride type, and buttons to navigate to other features of the dashboard.

³<https://www.microsoft.com/en-us/microsoft-365/p/excel/cfq7ttc0hr4r?activetab=pivot:overviewtab>

⁴<https://pandas.pydata.org/>

⁵This project used Visual Studio code with Javascript, CSS, HTML, and Copilot to create the dashboard. Link for Visual Studio Code <https://code.visualstudio.com/>

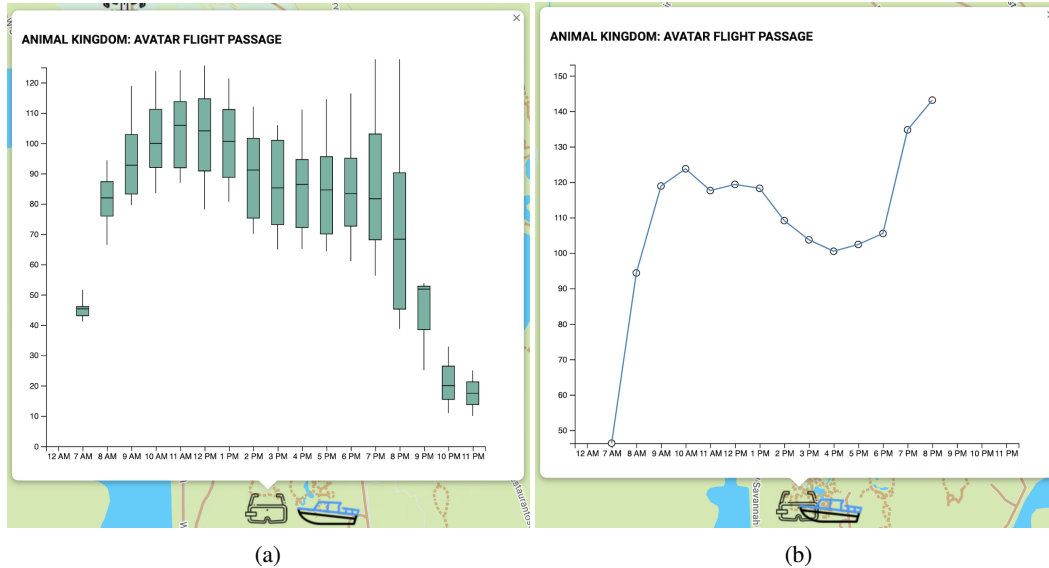


Figure 3: Wait times for Animal Kingdom's Avatar Flight of Passage.
 (a) Animal Kingdom's Avatar Flight of Passage overall wait time distribution. Peak time is late morning/early afternoon.
 (b) Animal Kingdom's Avatar Flight of Passage wait times in January. Peak time is in evening.

When a marker is clicked, the name of the corresponding attraction and the park in which the ride is located in appear on a popup window. Additionally, the popup window will have either an overall hourly wait time distribution or a monthly hourly wait time line plot, depending on if a month filter is on or not. This is shown in Figure 3.

When the Best Time button, which is located on the sidebar (see Figure 2 for the button location), is clicked, a new page will appear and show the slowest and busiest times of year at Walt Disney World. The busiest month overall at Disney World is December with a wait time of approximately 61 minutes. By park, the busiest month to go to Animal Kingdom is in February (average wait time of 61 minutes), the busiest month to go to Epcot is in December (average wait time of 68 minutes), the busiest month to go to Hollywood Studios is in February (average wait time of 72 minutes), and the busiest month to go to Magic Kingdom is in April (average wait time of 58 minutes). The quietest month overall at Disney World is September with a wait time of approximately 43 minutes. September is also the quietest month to go to Animal Kingdom average wait time of 40 minutes), Hollywood Studios (average wait time of 47 minutes), and Magic Kingdom (average wait time of 39 minutes), but the quietest month to visit Epcot is in May (average wait time of 48 minutes). Lastly, the holiday with the busiest wait time is New Year's Day with an average wait time of 74 minutes and the quietest holiday is Labor Day with an average wait time of 38 minutes.

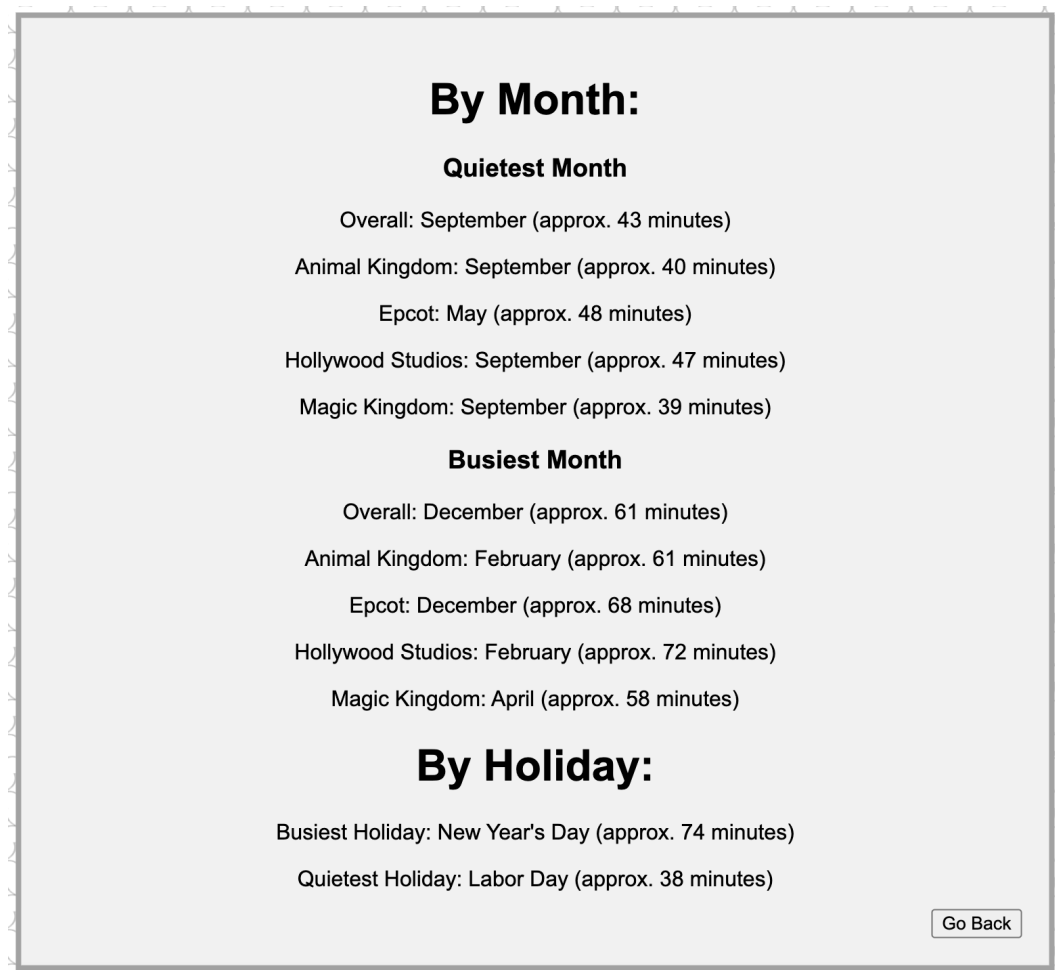


Figure 4: Best Times page on dashboard. Categorized by month and holiday, users can find the quietest and busiest of each category.

When the Compare Waits button, which is located on the sidebar (see Figure 2 for the button location), is clicked, users will be able to select a month and multiple rides to compare their hourly wait time. This is intended to help maximize the time at a park by planning when to go on which ride, but also to help when making reservations to restaurants, or if someone is going to multiple parks in one day, when to go to which park. See Figures 5 and 6 for examples.

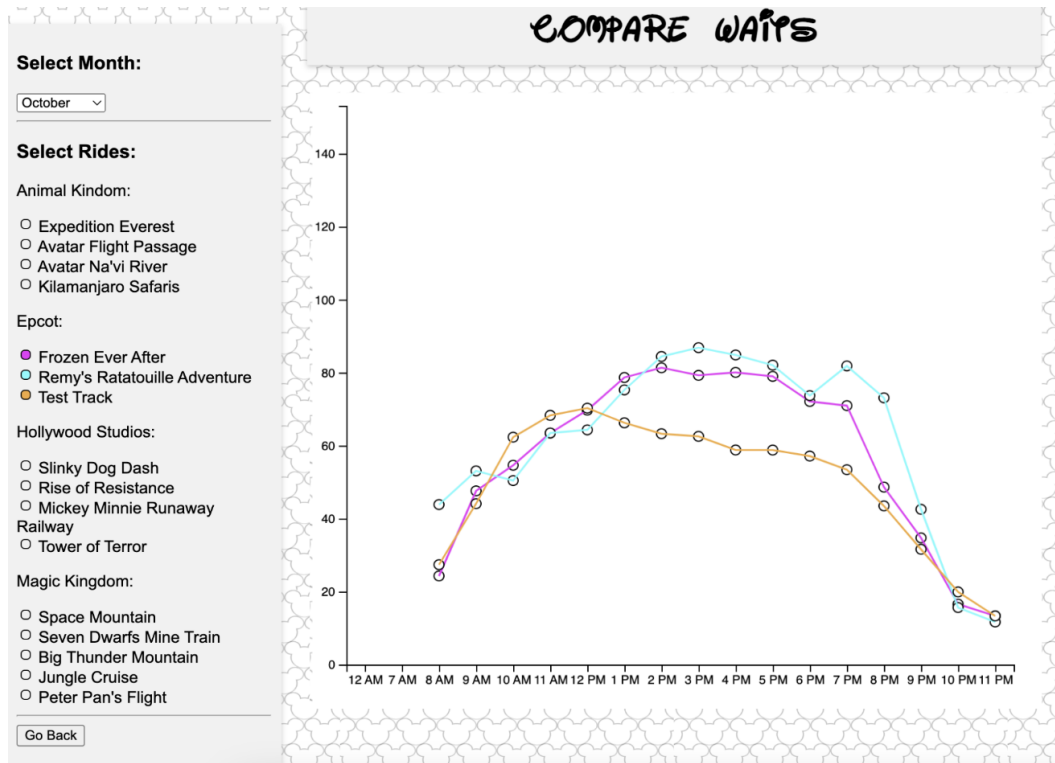


Figure 5: Comparing waits for Frozen Ever After, Remy's Ratatouille Adventure, and Test Track in October to find best time to go on each ride.

Figure 5 is comparing waits with attractions located at the same park. This plot suggests to plan on going on Frozen Ever After either around 8am or after 8pm, Remy's Ratatouille Adventure around 10am or after 8pm, and Test Track from 12PM to 8PM.

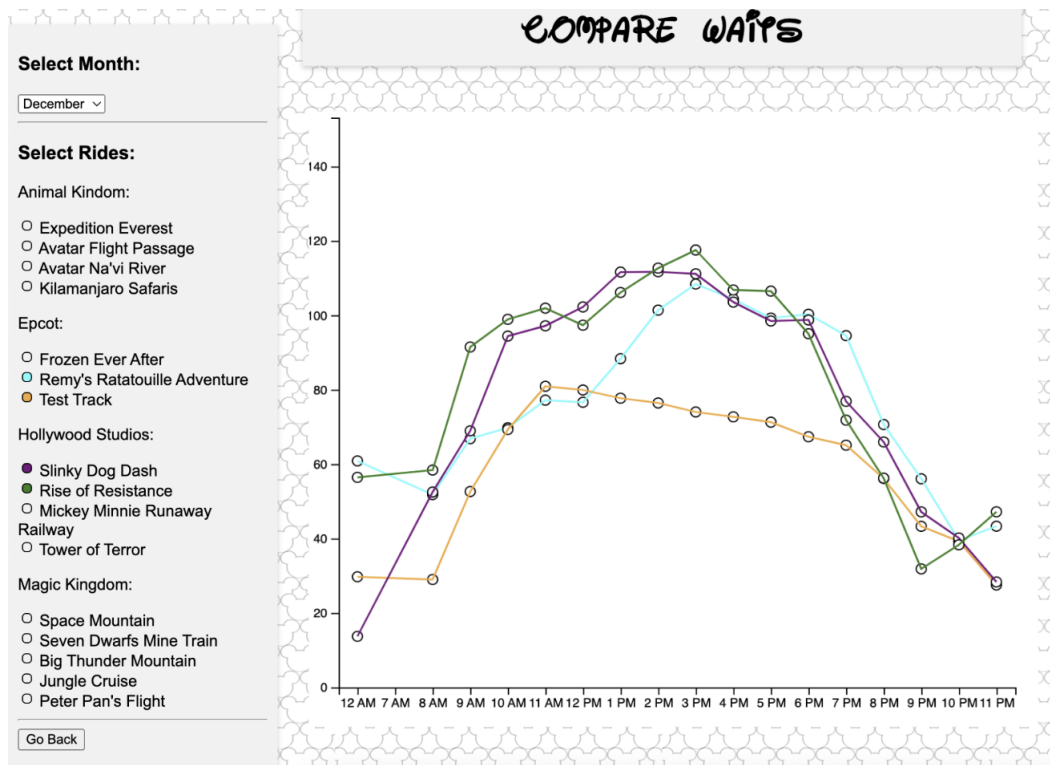


Figure 6: Comparing waits for Epcot's Remy's Ratatouille Adventure, Epcot's Test Track, Hollywood Studios' Slinky Dog Dash, and Hollywood Studios' Rise of Resistance in December to find best time to go on each ride and best time to go to each park.

Figure 6 is comparing waits with attractions located at two different parks: Epcot and Hollywood Studios. If a guest buys a park hopper ticket, this means that a guest can go to multiple parks in one day. ⁶ With this information, this plot suggests to plan on going to Epcot at the beginning of the day and Hollywood Studios at the end of the day.

⁶Park hopper information: <https://disneyworld.disney.go.com/guest-services/park-hopper/>

Future Work

The Disney Wait Time dashboard has a few limitations. First off, which was mentioned in the **Data** section, was that there are only 16 attractions included on this dashboard. While these are the attractions that tend to have longer waits, having more rides on this dashboard will improve the helpfulness of it. Additionally, the data used for this dashboard is only from February 2023 to January 2024, and data from previous years would also improve accuracy and a truer average wait time for each month. For example, Magic Kingdom's TRON Lightcycle Run⁷ attraction opened in April 2023, which could be why Magic Kingdom's busiest month was in April.

Lastly, rides that were closed during park hours did not have wait times included. As mentioned in **Related Work**, Hollywood Studios' Rise of the Resistance is closed fairly often. Closure times were not included in the data, which may lead to some inaccuracies in hourly wait time averages.

To improve dashboard functionality, besides adding more rides, ride closures can be recorded, park hours can be posted, and current wait times could be posted.

⁷<https://disneyworld.disney.go.com/attractions/magic-kingdom/tron-lightcycle-run/>

References

- [1] Suzanne Rowan Kelleher. Disney has 7 of the world's 10 most-visited theme parks, universal has other 3, Jun 2023.
- [2] Ady Milman, Asli D.A. Tasci, and Wei Wei. Crowded and popular: The two sides of the coin affecting theme-park experience, satisfaction, and loyalty. *Journal of Destination Marketing Management*, 18, 2020.
- [3] Luke Dammann. Disney world ride hits staggering 320-minute wait time, Jul 2023.
- [4] Maksim Godovykh, Ady Milman, and Asli Tasci. Theme park experience: Factors explaining amount of pleasure from a visit, time allocation for activities, perceived value, queuing quality, satisfaction, and loyalty. *Journal of Tourism and Leisure Studies*, 4:1–21, 01 2019.
- [5] Yingsha Zhang, Xiang Li, and Qin Su. Does spatial layout matter to theme park tourism carrying capacity? *Tourism Management*, 61:82–95, 02 2017.
- [6] Shizhen Bai, Hao He, Chunjia Han, Mu Yang, Dingyao Yu, and Xinrui Bi. Exploring thematic influences on theme park visitors' satisfaction: An empirical study on disneyland china. *Journal of Consumer Behaviour*, 23:90–106, 03 2023.
- [7] Henk Roest, Rik Pieters, and Kitty Koelemeijer. Satisfaction with amusement parks. *Annals of Tourism Research*, 24(4):1001–1005, 1997.
- [8] Gary Geissler and Conway Rucks. The overall theme park experience: A visitor satisfaction tracking study. *Journal of Vacation Marketing*, 17:127–138, 04 2011.
- [9] Ady Milman and Asli D.A. Tasci. Exploring the experiential and sociodemographic drivers of satisfaction and loyalty in the theme park context. *Journal of Destination Marketing Management*, 8:385–395, 2018.
- [10] Yanyan Zheng, Wei Wei, nathaniel Line, and Lu Zhang. Integrating the tourist gaze with the social servicescape: Implications for creating memorable theme park experiences. *International Journal of Hospitality Management*, 93, 12 2020.
- [11] Edwin Torres, Ady Milman, and Soona Park. Customer delight and outrage in theme parks: A roller coaster of emotions. *International Journal of Hospitality Tourism Administration*, 22:1–23, 07 2019.
- [12] Changhee Kim and Soowook Kim. Measuring the operational efficiency of individual theme park attractions. *SpringerPlus*, 5:1–9, 06 2016.
- [13] Hannah Risser. Rise of resistance broke down over 500 times last year in disney world, Aug 2023.